



My Personal Journal Through Artificial Intelligence

Final Project, John Zimmerman

AI Fundamentals

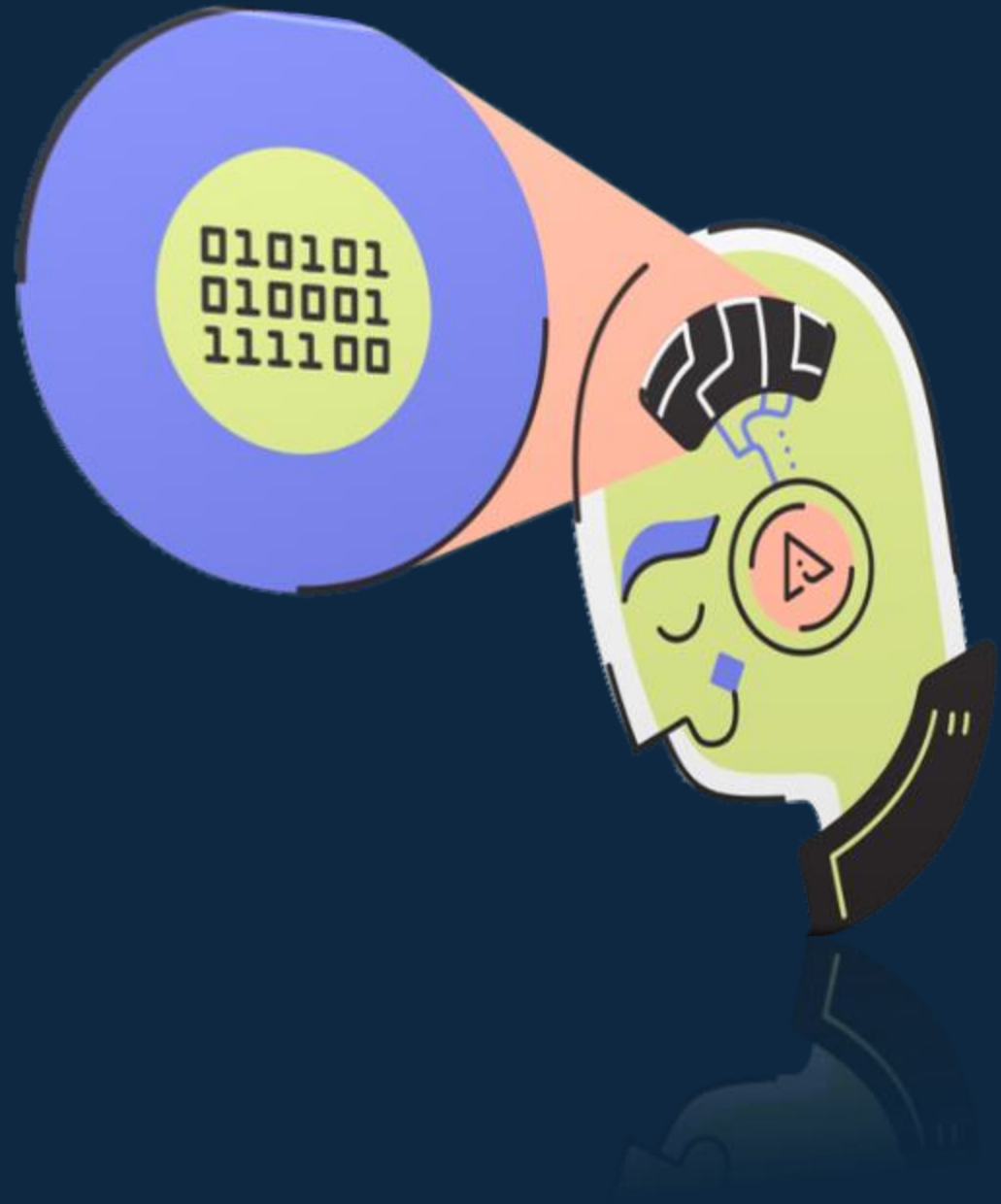
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The History of AI

REFLECTION

The most interesting part of this class regarding the history of AI was recognizing many of the pop culture examples that I grew up with. I grew up reading Neuromancer and watching the movie Hackers, so I sometimes feel like I should have been born in a different era. Another topic I found especially interesting was learning about Google's AlphaGo and AlphaFold projects. AlphaFold in particular, is one of the most impressive and exciting feats in artificial intelligence today. The first week I picked an article about the chess engine that beat the Grand Master Gary Kasparov, and it's interesting to see how far AI has come in that short amount of time since.



Machine Learning

REFLECTION

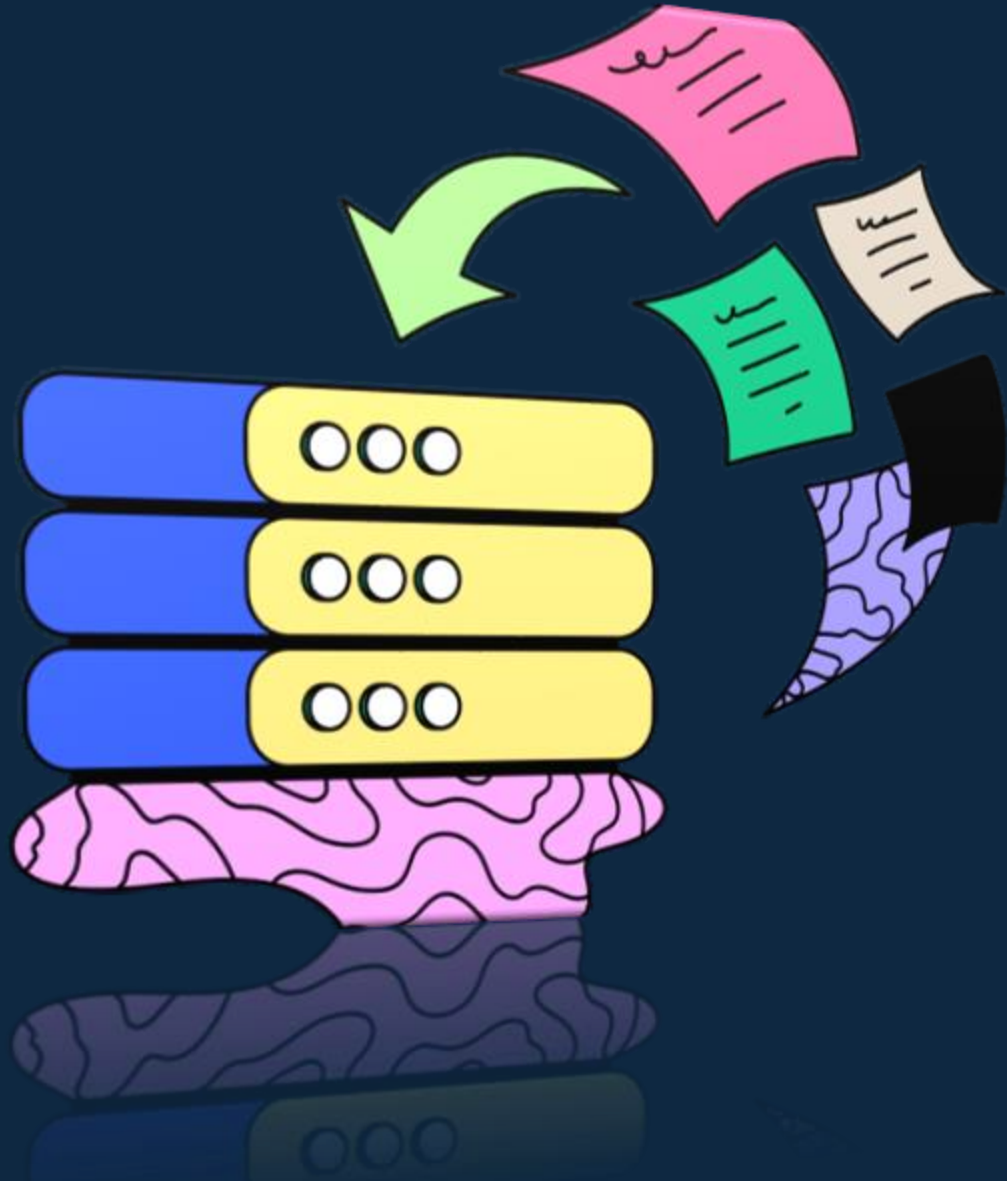
Machine Learning and learning about Neural Networks were some of the most interesting materials we got to look at all semester. Before starting the assignment to create a neural network diagram, I watched this YouTube video by the channel 3Blue1Brown, and it really explains the flow between the layers. The way it was described really helped me understand how input is passed through the layers and how each hidden layer is tasked with different things. I would say this was my favorite part of the class, as it felt like we actually got to dive into the technical information about AI.



Data Management

REFLECTION

During week 4, we looked at data management with AI, and the degree to which AI systems depend on high-quality data. Before this class, I mostly thought about AI in terms of algorithms and models, but it was interesting to learn that data collection and management are equally important. The concept of "garbage in, garbage out" makes clear that unreliable or biased data can directly affect the performance and trustworthiness of AI systems.



LLM's

REFLECTION

I enjoyed this week's section because it went in depth on how LLM's are applied in real-world situations. I found it especially interesting to learn more about transformer architecture, particularly the computational and environmental costs of training LLMs, such as energy consumption and the downstream effects large-scale GPU data centers have on local ecosystems. Also, I enjoyed diving into prompt engineering and liked doing the assignment that week. I actually used that assignment to learn about recursion using AI, and I was able to understand the topic better than I have in the past.



Ethical AI

REFLECTION

One of the more interesting parts of learning about ethical AI was how quickly artificial intelligence is beginning to affect fields that aren't directly tech jobs. The week we talked about "Ethical AI," I researched AI gene-editing technologies. Specifically, Open-CRISPR and how it can edit genetic structures that don't exist in nature. The benefits of this could dramatically expand the possibilities of treating incurable diseases currently. On the other hand, it raises ethical questions and issues of accessibility, data transparency, and corporate control, all of which become important when technologies have the potential to fundamentally change healthcare and human life.



Compromised AI

REFLECTION

For the week 6 topic, “Comprised AI,” I found an article on how autonomous systems could fundamentally change warfare and the ethical implications that arise from them. My discussion focused on AI-controlled military aircraft and the ethical concerns of autonomous weapons. I found it especially hard to weigh the costs and benefits because it's hard to quantify them, as real lives hang in the balance on both sides of the issue. The assignment made me consider how AI advancements are not only technical issues but also political and ethical ones that could reshape public attitudes towards conflict and war.



Future Concepts

REFLECTION

This week, we considered emerging technologies, and I specifically researched smart prosthetics. I found it interesting that the lecture slides discussed most major technological advancements, from agriculture to automation, and how there were costs and benefits to all the technologies we use today. My topic on smart prosthetics reflected this idea, because artificial Intelligence, sensor tech, and neural control systems could dramatically improve the quality of life for amputees, but concerns about privacy, ethics, and physical costs may outweigh those benefits in the short term. So even though these advancement set out to solve problems, they typically create new ones as well.



What I learned from This Course

1. How your understanding of AI changed

Like I said in a past slide, before this course I mostly thought of AI as chatbots and generative tools, but the material in the course, especially the slides, helped me understand how broad the field is. One thing I found especially interesting is how technologies that were once considered "AI" often stop being viewed that way once they become common or well understood. Things like recommendation systems, or basic automation now feel ordinary compared to modern generative tools.

2. Important concepts that you learned

I think one of the most important concepts I learned was developing a better understanding of how systems like neural networks and large language models actually function under the hood. Before this course, AI felt more like a magic box, and even if it still does to a certain degree, I at least feel confident in being able to understand more technical aspects of these things when listening to different educational material.

3. New skills that you developed

The most obvious new skill I learned is how to better prompt AI for educational purposes. It's become apparent that there is a certain way to go about interacting with AI to get the best information possible.

4. Anything to share about the structure and/or the progression of the course

One criticism I would give is that some of the more technical assignments, especially earlier in the course, felt underexplained during lectures. For example, when we learned about neural networks and had to create network diagrams, it sometimes felt like we were expected to independently research parts of the material rather than fully learning it through class instruction. Other than that, I thought the overall structure of the course was solid.

How I Envision Using AI in the Future

1. How you might use AI in the future

I will continue to use AI as I have to help me with learn. I have always used AI to help me write small quizzes and flash cards, as I feel like its most useful when you give it some information and ask it to make study resources.

2. How do you envision its impact on you and society

I envision AI's impact on society that it will change so much that it's hard to even guess what the world will look like. In the near future, I suspect that it will continue to change the economy in ways where the outcomes are hard to quantify.

3. Additional learning related to AI that you might explore

I bought a book on machine-learning and it goes over some things about TensorFlow and Pytorch. In the meantime, I need to get some of my math skills up to par, and then I can jump into it.

Thank you!

Feedback:

- Like I stated before, my favorite part was learning more about Neural Networks, and it would have interesting to learn more deeply on the technical aspects of AI.
- The slides were helpful and engaging for the most part. If I had to choose, I wish we covered more of the earlier technical stuff about Neural Networks, Fuzzy Logic, Machine Learning, Heuristics, etc. And then the other thing that would have been cool to go in depth would be on prompting.